

Project: Customer Purchase Behavior (Decision Tree)

Objective: Explainable Classification.

Method: Decision Tree with Class Balancing.

Created by: AI Agent (Antigravity)

Problem Statement

Problem: 'Black Box' models are hard to trust.

Goal: Explain WHY a customer is classified as 'Sports'.

Context: Regulatory environments often mandate explainability.

Real-World Use Case

Scenario: Loan Approvals / Fraud.

Reason Code: 'Denied because $\text{Income} < X$ and $\text{Debt} > Y$ '.

E-commerce: Debugging why a campaign failed.

Input Data

Features: Standard (Age, Income, etc.).

Scaling: Not strictly needed for Trees but used for consistency.

Weights: Using 'balanced' weights to help minority classes.

Concepts Used

1. Nodes/Leaves: Questions and Answers.
2. Gini Impurity: Metric for splitting.
3. Max Depth: Pruning to prevent overfitting.
4. Interpretability: White-box modeling.

Concept: 20 Questions

Think of a Decision Tree as a game of 20 Questions.

Q1: Is spending > 500? (Yes/No)

Q2: Is Age > 30? (Yes/No)

Answer: Electronics Category.

Step-by-Step Flow

Step 1: Load & Clean.

Step 2: Configure Tree (max_depth=4, class_weight='balanced').

Step 3: Train.

Step 4: Visualise Logic.

Step 5: Evaluate.

Code Logic Summary

Model: `DecisionTreeClassifier(max_depth=4, class_weight='balanced')`.

Visualization: `plot_tree()`.

Execution Results

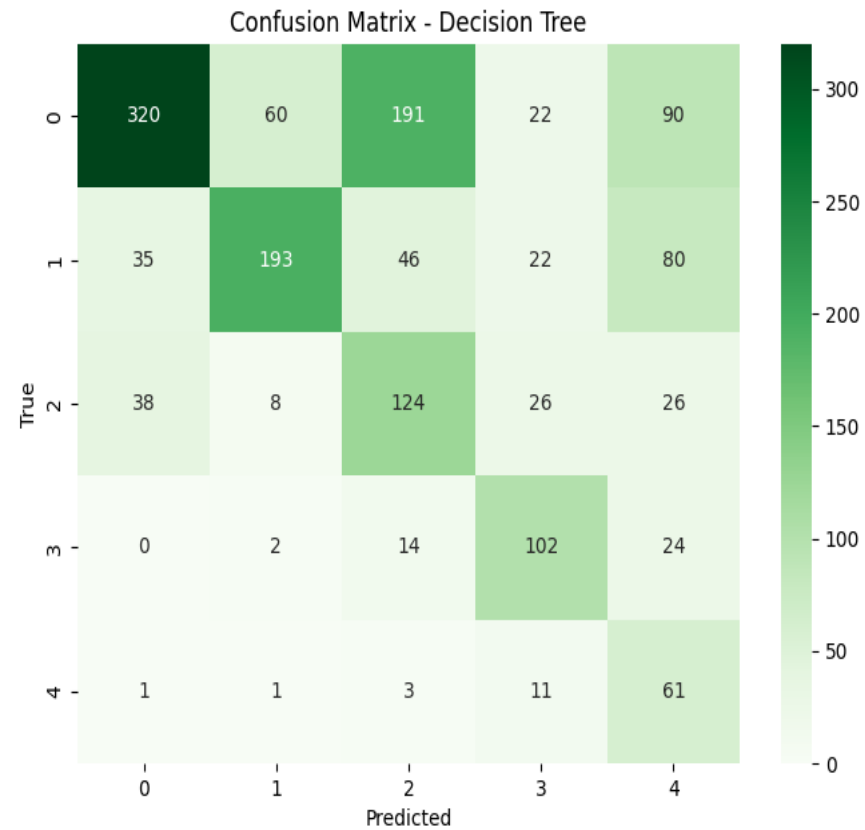
Overall Accuracy: (Check Code Output).

Recall: Should be better for Class 4 (Sports) due to weights.

Trade-off: Lower precision potentially.

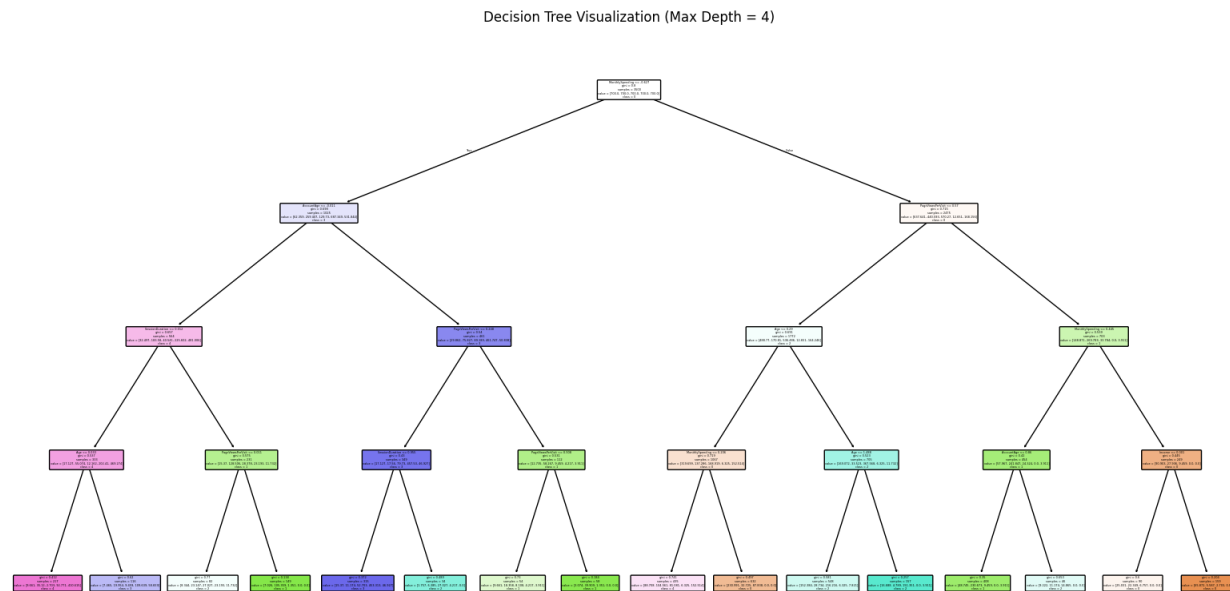
Confusion Matrix

Visualizing performance:



Tree Visualization

The logic map:



Advantages & Limitations

Pros: Explainable, handles non-linearities, no scaling needed.

Cons: Overfitting (if deep), Instability (high variance).

Interview Corner

Q: Gini vs Entropy? A: Similar results, Gini is faster.

Q: Pruning? A: Limiting depth to improve generalization.

Conclusion

Summary: We have an explainable model.

Next: Random Forest to average out the variance of a single tree.